The Indian EV market was valued at just US$5 billion in 2020 with EV sales barely accounting for 1.3% of the total vehicles sold between 2020 and 2021. However, the market is projected to reach US$46 billion by 2026 with a CAGR of 44% between 2021 and 2026. An even higher growth of US$150 billion is being projected between 2021 to 2030 with a CAGR of 90%.

Therefore, India offers a nascent, untapped EV market that continues to draw a lot of attention with many vendors — both startups and biggies — vying for a slice of the average commuter's bread. But is India capable of meeting the surging EV demand? Let's take a brief look at some of the challenges that lie ahead in the electrification of the Indian automobile sector.

RANGE ANXIETY

Ask a prospective EV buyer about the single biggest concern they have about switching to EV, and 9/10 times the answer is likely to be the range of the vehicle and the fear of running out of power in the middle of a journey. This is not without reason, though, and the availability of multiple range testing standards with varying accuracies only serves to further complicate the situation.

The range of any EV is assessed based on several standards including New European Driving Cycle (NEDC), Worldwide Harmonized Light Vehicle Test Procedure (WLTP), which are both European, and the Environmental Protection Agency (EPA) test, which is a US standard. The NEDC and WLTP focus more on city-based driving estimates while EPA focuses on highway driving.

In India, EV testing and certification is done by the Automotive Research Association of India (ARAI). ARAI has stringent ideal testing conditions, which do serve as benchmarks, but on-road conditions are likely to be vastly different. Besides, independent testing has shown that the attainable range on-road is likely to be different than advertised ranges due to various factors beyond the user's control.

ADDRESSING RANGE ANXIETY

Mitigating range anxiety is important for companies to instill a sense of confidence in consumers and increase penetrance of EVs in the market. Increasing battery efficiency and expansion of charging networks is one way to solve the problem. Development of enhanced navigation systems with better understanding of traffic along with indication of charging points and bypass routes can help commuters make the most of available charge in reaching their destinations. Intelligent systems in the vehicle that pre-condition the battery enroute to a charging station.

CHARGING INFRASTRUCTURE

Lack of adequate charging infrastructure contributes to range anxiety. It is only lately that Central and State governments have been encouraging stakeholders to strengthen EV charging infrastructure.

According to the Charging Infrastructure (CI) for EV policy released by the Ministry of Power in 2019, the regulatory framework is comprised of three main bodies. These include: the Bureau of Energy Efficiency (BEE), which acts as the Central nodal agency, the State electricity distribution companies or 'DISCOMs' that act as the nodal agencies for corresponding state governments, and an Implementation Agency that coordinates between the two aforementioned nodal agencies and is responsible for installation of charging stations.

India currently has about 1,742 charging stations as of March 2022, with the bulk of these being concentrated in nine major cities such as Delhi, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad, Pune, and Surat. Compare this to China's 456,000 charging stations in 2018 alone and you can clearly see there's a lot of catching up to do.

Land acquisition, lack of clarity in tariffs, concerns in managing added electricity load, and overall high costs of electricity itself are some of the challenges that impede widespread installation of EV charging points. The lack of affordable renewable energy also puts a question of the sustainability of these charging points on the existing coal power grid. There's also a severe dearth of open parking spaces in India, which further disincentivizes people to own EVs.

Compounding the lack of charging facilities is the long times it takes to fully charge an EV. Even with a fast 300 kW charger, it takes no less than 20 full minutes to bring a car to 100% capacity with fast charging not to mention the plethora of charger connector types for each manufacturer.

DEVELOPING CHARGING INFRASTRUCTURE

The CI policy envisions adequate EV charging infrastructure in in the first three years in mega cities with